IED 1.6 | Measurement, Precision & Tolerances

Measurement & Inspection Record

NAME: ________________________________ PERIOD: __________ DATE: ________________________________

DAY 1 | MEASUREMENT & PRECISION

Build reliable measurement habits

Your goal is to select an appropriate measuring tool, record a realistic value, and check whether your measurement is repeatable.

Tool resolution: Ruler __________ Caliper __________

Feature	Tool	Resolution	Trial 1	Trial 2	Trial 3
Overall length					
Overall width					
Thickness / height					
Outside feature					
Hole / inside feature					

REPEATABILITY CHECK

Selected feature: ______________________________

Smallest value: __________ Largest value: __________ Range = largest - smallest: __________

What could cause your repeated measurements to differ?

____

What did you do to improve consistency?

____

Which tool gave you the most confidence for this feature, and why?

____

NAME: ______________________ PERIOD: ________ DATE: ______________________

WARM-UP | NOMINAL, TOLERANCE & LIMITS

For a bilateral tolerance, the lower limit is nominal - tolerance and the upper limit is nominal + tolerance. A measured feature passes when it is inside or exactly on the limits.

Example: 0.500 +/- 0.005 in

Lower = 0.495 in Upper = 0.505 in Acceptable range: 0.495 through 0.505 in

Nominal +/- Tolerance	Lower Limit	Upper Limit	Measured Value	Pass / Fail
1.000 +/- .020			0.984	
.500 +/- .010			.509	
Ø.500 +/- .005			.506	
1.500 +/- .020			1.480	
.090 +/- .005			.087	

BEFORE YOU INSPECT

1. Read the entire drawing before measuring.
2. Calculate limits before comparing an actual part.
3. Measure the feature shown by the dimension - not a nearby surface.
4. Record the value with precision appropriate to the tool.
5. A single failed required feature causes the part to be rejected.

NAME: ______________________ PERIOD: __________ DATE: ______________________

MODEL A | THROUGH HOLE BLOCK

Inspect size and hole location

Feature	Nominal +/- Tol.	Lower	Upper	Actual	P/F
Overall length	1.600 +/- .020				
Overall height	1.000 +/- .020				
Overall depth	1.000 +/- .020				
Hole diameter	Ø.500 +/- .005				
Hole X location	.800 +/- .010				
Hole Y location	.500 +/- .010				

MODEL A DISPOSITION: ACCEPT / REJECT **Failed feature(s):** ______________________

MODEL B | TWO LEVEL STAIR BLOCK

Inspect overall and stepped geometry

Feature	Nominal +/- Tol.	Lower	Upper	Actual	P/F
Overall length	1.500 +/- .020				
Overall height	.750 +/- .020				
Overall depth	1.000 +/- .020				
First step run	.500 +/- .010				
Second run / location	1.000 +/- .010				
First step height	.250 +/- .010				
Second step height	.500 +/- .010				

MODEL B DISPOSITION: ACCEPT / REJECT **Failed feature(s):** ______________________

NAME: ______________________ PERIOD: ________ DATE: ______________________

MODEL C | SLOTTED BRACKET

Inspect multiple feature types

Feature	Nominal +/- Tol.	Lower	Upper	Actual	P/F
Overall length	1.500 +/- .020				
Overall height	1.500 +/- .020				
Overall depth	1.500 +/- .020				
Hole diameter	Ø.400 +/- .005				
Hole X location	.400 +/- .010				
Hole Y location	.400 +/- .010				
Slot width	.500 +/- .010				
Slot location / length	1.000 +/- .010				
Thin wall / web	.090 +/- .005				

MODEL C DISPOSITION: ACCEPT / REJECT **Failed feature(s):** ______________________

ENGINEERING ANALYSIS

Which inspected dimension had the tightest tolerance? Why might it require tighter control?

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Why can a part look correct but still fail inspection?

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Choose one rejected feature. What manufacturing change could move it closer to nominal?

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